rag -- 2026-11-07

5 new paper(s) · topic: Retrieval-augmented generation: retrieval quality, grounding and faithfulness, evaluation benchmarks, and RAG system design.

Worth reading in full

[P2] They Know Where It Is, But Not What It Says: A Bilingual Hallucination Benchmark of Hadith Recall in Open LLMs -- It provides a rigorous, bilingual hallucination benchmark that quantifies exact recall failures of open-weight models, directly addressing grounding and faithfulness concerns central to retrieval-augmented generation.

Themes

These papers converge on the central role of retrieval mechanisms for grounding generated text. AssameseLM-E2B demonstrates that a byte-level vocabulary and parameter-efficient continual pre-training can boost RAG scores to 96.7 % while cutting token-per-word ratios, highlighting the impact of retrieval-aware fine-tuning on low-resource generation [P1]. Vectorium and kANNolo take a complementary angle by providing a modular ANN library that supports dense, sparse and multi-vector indexes without runtime overhead, achieving state-of-the-art accuracy-throughput trade-offs on SIFT1M and Dragon collections, which directly benefits the retrieval back-ends used by RAG systems [P4]. Both works therefore address retrieval quality from opposite ends of the stack - model-side augmentation versus index-side efficiency.

Grounding and faithfulness are examined from the perspective of hallucination risk. The Hadith recall benchmark reveals that open-weight LLMs (3.8-8 B) fail to reproduce exact source texts, with 0 % exact matches and hallucination rates up to 75 % despite bilingual prompting, underscoring severe factuality gaps in generation without retrieval support [P2]. AssameseLM-E2B shows a bimodal refusal bias and variable factuality scores [P1]. Together, these studies expose a shared evaluation gap: most works lack a unified metric that jointly measures retrieval precision and downstream factual consistency.

Evaluation benchmarks and system design are further explored in the cultural-translation and AEC domains. CULTURE-CMT introduces a retrieval-augmented judging model (Qwen3.5-27B) to score cultural effectiveness, finding that context improves scores modestly (+0.096) but larger models or chain-of-thought inference do not guarantee gains, highlighting the nuanced interaction between retrieval cues and model scaling [P5]. The AEC review proposes readiness criteria (machine-readability, provenance, etc.) and notes that while RAG and the Model Context Protocol can address format-level interoperability, semantic and organizational challenges remain largely untested, pointing to a need for concrete benchmarks akin to those in P2 and P5 [P3].

Overall, the set reveals a split focus: P1, P4, and P5 advance retrieval infrastructure and context usage, whereas P2 and P3 diagnose grounding failures and interoperability gaps. The recurring absence of end-to-end, task-specific benchmarks that combine retrieval accuracy, factual grounding, and domain-specific constraints is the most salient shared limitation.

6 theme sentence(s) checked against the paper summaries; 1 corrected, 0 removed.

At a glance

[P1] AssameseLM-E2B: An Open-Weight Assamese Language Model for Efficient Consumer-Grade Deployment · relevance 0.31
AssameseLM-E2B is a 2-B parameter, byte-level-expanded Gemma-4 model that achieves low-resource Assamese performance (BPB 0.7488, RAG 96.7 %) through parameter-efficient continual pre-training and instruction fine-tuning, but shows script-mixing issues and a bimodal refusal bias.

[P2] They Know Where It Is, But Not What It Says: A Bilingual Hallucination Benchmark of Hadith Recall in Open LLMs · relevance 0.28
Open-weight LLMs (3.8-8 B parameters) cannot reliably reproduce exact Hadith text from a bibliographic reference, achieving 0 % exact recall across 30 000 attempts.

[P3] Toward Agentic AI-Ready Workflows in Built Environment Organizations: Knowledge Structures and Interoperability Challenges · relevance 0.28
The paper reviews how heterogeneous built-environment data must be organized and made interoperable to enable reliable agentic AI workflows, highlighting current fragmentation and proposing readiness criteria and research directions.

[P4] Modular and Efficient Approximate Nearest Neighbors Search with Vectorium and kANNolo · relevance 0.25
Vectorium and kANNolo are modular Rust libraries that achieve state-of-the-art ANN performance across dense, sparse and multi-vector data with no runtime overhead.

[P5] CULTURE-CMT: Benchmarking Contextualized Social Media Translation · relevance 0.24
CULTURE-CMT is a 572-example Chinese-English benchmark that evaluates social-media comment translation with context and cultural effectiveness, revealing that context helps but larger models and longer reasoning do not uniformly improve performance.

[P1] AssameseLM-E2B: An Open-Weight Assamese Language Model for Efficient Consumer-Grade Deployment

Pragyan Phukan · arXiv 2611.06170v1 · https://arxiv.org/abs/2611.06170v1 · relevance 0.31

TL;DR: AssameseLM-E2B is a 2-B parameter, byte-level-expanded Gemma-4 model that achieves low-resource Assamese performance (BPB 0.7488, RAG 96.7 %) through parameter-efficient continual pre-training and instruction fine-tuning, but shows script-mixing issues and a bimodal refusal bias.

Problem
Large language models often underperform on low-resource languages because their tokenizers cause severe subword fragmentation, generative instability, and linguistic interference, especially for scripts like Eastern Nagari used in Assamese [1][2].

Method
The authors build AssameseLM-E2B on the 2-B Gemma-4-E2B backbone, expanding the byte-level BPE vocabulary to cut tokenization overhead by roughly 500 % versus multilingual 2-3 B baselines [1][2]. They apply a parameter-efficient continual pre-training (CPT) pipeline using Hybrid LoRA, followed by supervised fine-tuning (SFT) on a balanced multi-domain dataset to improve instruction following and retrieval-augmented generation (RAG) [1][3][4]. Training uses a cosine learning-rate schedule peaking at 2.0 × 10?4, with CPT run for 0.5 B tokens and SFT for ~1400 steps [3][4].

Key results
- Bits-per-Byte (BPB) on FLORES-200: 0.7488 [1].
- Overall RAG score: 96.7 % and 100 % refusal accuracy on adversarial unanswerable prompts [1].
- Token-per-word ratio improved from 2.7 (baseline) to 1.5 after vocabulary expansion [2].
- CPT dynamics: final cross-entropy loss ~ 3.1, perplexity stabilized at 101-102, next-token accuracy plateaued just above 40 %, and prediction entropy converged to ~ 3.0 [3].
- SFT loss EMA settled between 2.5-3.5, gradient-norm EMA decayed to ~ 0.5 (despite occasional spikes) [4].
- Instruction-following evaluation (four dimensions):
* Linguistic Accuracy mean 7.30 (vs. Llama-3.2-1B-Instruct 1.26, Qwen-3-3B-Instruct 1.31, Gemma-4-E2B-it 3.53) [5].
* Query Understanding mean 8.06 (vs. Llama 6.48, Gemma 7.47, Qwen 2.00) [5].
* Factuality mean 4.89 and Instruction Following mean 4.62, both showing a bimodal "hourglass" distribution [5].

* Baseline models' Instruction Following scores were flat-lined at 1.40 [5].

Limitations
The model exhibits generative degradation on Romanized Assamese and unintended assimilation of Bengali script, reflecting residual tokenizer and script-mixing issues [1]. Evaluation reveals a bimodal refusal bias: the model either answers correctly (high scores) or issues a canned refusal (low scores) [5]. Training shows high variance in SFT loss and occasional gradient-norm spikes, indicating instability on complex instruction batches [4]. The paper's results are limited to FLORES-200, RAG benchmarks, and a few instruction-following metrics; broader task coverage is not reported.

Sources

[1] arXiv 2611.06170v1.pdf, arXiv abstract

[2] arXiv 2611.06170v1.pdf, page 4

[3] arXiv 2611.06170v1.pdf, page 8

[4] arXiv 2611.06170v1.pdf, page 12

[5] arXiv 2611.06170v1.pdf, page 16

[P2] They Know Where It Is, But Not What It Says: A Bilingual Hallucination Benchmark of Hadith Recall in Open LLMs

Dayam Nadeem, Safdar Mustafa, Ameen Ur Rehman, Shahab Saquib Sohail et al. · arXiv 2611.02587v1 · https://arxiv.org/abs/2611.02587v1 · relevance 0.28

TL;DR: Open-weight LLMs (3.8-8 B parameters) cannot reliably reproduce exact Hadith text from a bibliographic reference, achieving 0 % exact recall across 30 000 attempts.

Problem
The paper addresses the high-stakes task of verbatim Hadith recall, where a single correct text is tied to a specific reference and any deviation constitutes a dangerous fabrication (e.g., misquoting the Prophet) [1][2].

Method
The authors build a bilingual (English/Arabic) benchmark covering all six canonical Sunni Hadith collections, sampling 500 Hadith per collection (3 000 responses per model per language, 30 000 total) [1][2]. Five locally-deployable open-weight models (Mistral-7B, Gemma3-4B, Jais-7B, ALLaM-7B, Phi4-mini-3.8B) are prompted only with the bibliographic reference, without any retrieval augmentation [1][2]. Responses are processed through a pipeline that first detects refusals, then computes lexical fidelity (sentence-level BLEU) and semantic fidelity (multilingual cosine similarity), and finally classifies each output into correct, theme-only, partial, hallucinated, or refused [3][4][5].

Key results
- Across all models and both languages, 0 % of the 30 000 responses achieve both high lexical and high semantic fidelity, i.e., an exact match to the source text [1][5].
- Average BLEU scores are near zero for both languages (English 0.005-0.008, Arabic 0.006-0.024) [5].
- Semantic cosine scores range from 0.447 to 0.736, with Arabic responses generally higher on cosine but still far below the correctness threshold [5].
- The proportion of "hallucinated" outputs dominates (e.g., 75.03 % hallucinated for Mistral-7B English, 32.27 % hallucinated for Mistral-7B Arabic) [5].
- Refusal rates are negligible (<=0.7 % in every cell) [5].
- Arabic responses fare uniformly worse than English in lexical fidelity, and Arabic-specialized pretraining does not improve exact recall [5][6].

Limitations
The study uses a publicly available, open-source Hadith dataset that, while verified for structural consistency, is a preliminary release and may contain residual noise [7]. Classification thresholds were set manually rather than learned, though the authors note the qualitative findings are robust to reasonable variations [7]. Evaluation is limited to five open-weight models of 3.8-8 B parameters run on consumer hardware; larger closed-source models are not assessed [7]. Finally, the benchmark evaluates zero-shot, retrieval-free recall, leaving the impact of retrieval augmentation to future work [7].

Sources

[1] arXiv 2611.02587v1.pdf, arXiv abstract

[2] arXiv 2611.02587v1.pdf, page 1

[3] arXiv 2611.02587v1.pdf, page 3

[4] arXiv 2611.02587v1.pdf, page 4

[5] arXiv 2611.02587v1.pdf, page 5

[6] arXiv 2611.02587v1.pdf, page 7

[7] arXiv 2611.02587v1.pdf, page 8

[P3] Toward Agentic AI-Ready Workflows in Built Environment Organizations: Knowledge Structures and Interoperability Challenges

Ahmet Bahaddin Ersoz, Justin McHenry, Yevgeniya Kovalchuk, Edward Barrett et al. · arXiv 2611.02286v1 · https://arxiv.org/abs/2611.02286v1 · relevance 0.28

TL;DR: The paper reviews how heterogeneous built-environment data must be organized and made interoperable to enable reliable agentic AI workflows, highlighting current fragmentation and proposing readiness criteria and research directions.

Problem
The transition to agentic AI in Architecture, Engineering, and Construction (AEC) hinges on data readiness, yet implemented systems frequently fail at the data boundary, with hallucinations and integration barriers limiting deployment [2]. Fragmentation and closed solutions dominate existing implementations, leading to isolated interoperability that does not extend across projects or organizations [5].

Method
The authors conduct a literature-based review, synthesizing findings into a three-layer framework that examines (1) the built-environment data landscape, (2) knowledge-structure and access mechanisms, and (3) interoperability challenges [3][4][5]. They identify five conditions indicating an information source's readiness for reliable agentic use: machine-readability, semantic explicitness, provenance/versioning, granularity, and update cadence [3]. The review also categorizes structural families--schemas, ontologies/graphs, and vector representations--and evaluates access mechanisms such as retrieval-augmented generation (RAG) and the Model Context Protocol (MCP) [4].

Key results
- Structured AEC data (e.g., BIM, CAD, schedules) are schema-defined but semantically incomplete, while unstructured data hold most decision-relevant knowledge yet lack explicit links [3][6].
- Six readiness conditions are proposed to assess data suitability for agents [3].
- Three structural families serve distinct agent operations: schemas for precise retrieval, graphs for relational reasoning, and vectors for semantic recall [4].
- Interoperability challenges are split into format-level (largely addressed), semantic (addressed only locally), and organizational (barely addressed) [5].
- The MCP tackles format-level and partly semantic interoperability but remains the least mature mechanism [4][5].

- No quantitative benchmarks, scores, or performance gains are reported; the paper calls for measurable criteria, deployment studies, and defined governance roles [6][7].

Limitations
The review notes a lack of shared benchmarks to independently test agentic readiness claims [6]. Governance mechanisms for permissioning, liability, and oversight are underdeveloped [6]. Organizational interoperability and a sector-wide role for maintaining semantic links are not addressed in existing studies [5][6]. The MCP, while promising, is identified as the least mature access mechanism [4][5].

Sources

[2] arXiv 2611.02286v1.pdf, page 2

[3] arXiv 2611.02286v1.pdf, page 4

[4] arXiv 2611.02286v1.pdf, page 6

[5] arXiv 2611.02286v1.pdf, page 8

[6] arXiv 2611.02286v1.pdf, page 10

[7] arXiv 2611.02286v1.pdf, page 12

[P4] Modular and Efficient Approximate Nearest Neighbors Search with Vectorium and kANNolo

Silvio Martinico, Franco Maria Nardini, Cosimo Rulli, Rossano Venturini · arXiv 2611.03927v1 · https://arxiv.org/abs/2611.03927v1 · relevance 0.25

TL;DR: Vectorium and kANNolo are modular Rust libraries that achieve state-of-the-art ANN performance across dense, sparse and multi-vector data with no runtime overhead.

Problem
Modern retrieval systems rely on Approximate Nearest Neighbors (ANN) over neural embeddings that can be dense, sparse or multi-vector, but existing ANN libraries fix a single combination of index structure, similarity function and quantizer, or only address one side of the problem, leaving many representation-index trade-offs unsupported [1].

Method
The authors introduce two open-source Rust libraries: Vectorium, a data-management layer that abstracts vectors, distance functions, quantizers and datasets independently and supplies 11 encoders ranging from 32-bit floats to sub-bit codes; and kANNolo, an indexing layer that implements generic HNSW and IVF indexes and a two-stage index that can combine any first-stage index with any rescoring dataset. Because every combination is specialized at compile time, the modular design adds no runtime overhead [1].

Key results
- Product Quantization (PQ) in Vectorium matches Faiss accuracy while offering competitive speed, and RaBitQ and HNSW implementations match or outperform their reference versions; IVF also outperforms Faiss [1].
On the dense SIFT1M collection, Vectorium's PQ achieves 32.1% Accuracy@10 versus 31.7% for Faiss, with query-per-second (QPS) of 9.4 versus 6.0 [3].
- On the dense Dragon collection (d=768, IP), Accuracy@10 is 53.8 % (both) with QPS 55.3 vs 37.2, and at larger scales (n=96 M) Accuracy@10 reaches 68.6 % (ours) vs 69.0 % (Faiss) with QPS 81.1 vs 65.0 [3].
- Compared to the reference RABITQ implementation, Vectorium's RABITQ yields up to 0.80 percentage-point accuracy difference (within run-to-run variance) but consistently smaller footprint (8-16 bytes less per vector) and higher throughput (1.44-1.74× on SIFT1M, 1.44-2.37× on Dragon) [3][5].
- Across seven public dense collections, the Vectorium/kANNolo indexes obtain the best memory-throughput trade-off in 9 of 12 collection-accuracy pairs and perform well on out-of-distribution

queries [1].
- For sparse data, systems built on Vectorium lead on all three collections; Seismic and SeismicWave are the fastest, while kANNolo's indexes are the most compact [1][6].
- Memory usage is the lowest among all compared systems: e.g., on MSMARCO-SPLADE-V3 the indexes require 6.8-7.6 GiB versus 7.9-10.6 GiB for SEISMIC and up to 30.1 GiB for LSP; on NQ they need only 2.0-2.1 GiB versus 3.4-8.7 GiB for others [6].
- Build times are longer for graph-based indexes (481 s to 5 428 s) than for inverted indexes (seconds to minutes), with the fastest builds belonging to SINDI (26-104 s) and SEISMIC (63-156 s) [6].

Limitations
- kANNolo incurs higher per-candidate scoring cost than a reference that uses a fast 1-bit first-pass filter, making the reference faster once candidate processing dominates (e.g., at nprobe >= 16 on SIFT1M) [5].
- Graph indexes built by kANNolo require substantially longer construction time than inverted indexes, which may be a bottleneck for very large collections [6].
- Some competing systems (e.g., PYANN) overflow 32-bit size counters on large sparse collections, causing failures, and LSP is not robust on many queries [6].

Sources

[1] arXiv 2611.03927v1.pdf, arXiv abstract

[3] arXiv 2611.03927v1.pdf, page 12

[5] arXiv 2611.03927v1.pdf, page 22

[6] arXiv 2611.03927v1.pdf, page 28

[P5] CULTURE-CMT: Benchmarking Contextualized Social Media Translation

Linjuan Wu, Dingming Li, Yiwen Qiu, Xinze Lyu et al. · *arXiv 2611.02018v1* · *https://arxiv.org/abs/2611.02018v1* · *relevance 0.24*

TL;DR: CULTURE-CMT is a 572-example Chinese-English benchmark that evaluates social-media comment translation with context and cultural effectiveness, revealing that context helps but larger models and longer reasoning do not uniformly improve performance.

Problem
Social-media comments are interactive and often contain slang or culturally specific expressions, yet existing translation evaluations treat them as isolated sentences, leading to mismatches when meaning depends on surrounding discussion or cultural knowledge[1].

Method
The authors build CULTURE-CMT, a benchmark of 572 manually verified comment translations organized into Slang, Context, and Both subsets to diagnose distinct sources of difficulty[1]. Each translation is scored on a four-level cultural-effectiveness rubric (0-3) that assesses literal meaning, affect, and communicative intent[2]. To enable scalable evaluation they develop a retrieval-augmented cultural-effectiveness Judger based on Qwen3.5-27B, fine-tuned with the Relax framework and trained to align with human scores[8]. Translation candidates are generated with prompts that optionally include the post and parent-comment context[7], and difficulty verification checks whether differences between independent translations reflect genuine challenges[7].

Key results
- Providing interaction context improves the mean cultural-effectiveness score for most models; e.g., Qwen3.8-Max's mean rises from 2.241 (no context) to 2.337 (with context), a gain of +0.096[3].
- The "Both" subset (requiring slang + context) remains the hardest, with lower scores across configurations[1].
- Model size does not correlate monotonically with performance; some larger models (e.g., Qwen3.5-397B) achieve lower mean scores than smaller ones[3].

- "Thinking" inference (chain-of-thought style) yields variable gains: Qwen3.6-27B-SFT (ours) improves mean from 2.241 to 2.365 (+0.124) with context, while other models see modest or negative changes[3].
- A 27 B model adapted on benchmark-disjoint social-media data attains the lowest observed invalidity rate under thinking inference with context, indicating targeted adaptation can reduce unusable outputs[1].
Invalidity rates for Qwen3.5-27B are 0.2 % both with and without context in thinking mode, and its mean score is 2.320 in both conditions [3].

Limitations
The study focuses exclusively on Chinese-English social-media comments, so findings may not generalize to other language pairs or domains[1]. The benchmark's diagnostic subsets (Slang, Context, Both) capture only three difficulty axes, leaving other possible sources of translation error unexamined[1]. While the Judger is trained on Qwen3.5-27B, the paper does not report results for alternative Judger architectures or for zero-shot cultural-effectiveness assessment[8]. Finally, the authors note that larger models are not consistently better and that "thinking" can sometimes degrade performance, indicating that scaling alone does not solve the problem[1].

Sources

[1] arXiv 2611.02018v1.pdf, arXiv abstract

[2] arXiv 2611.02018v1.pdf, page 4

[3] arXiv 2611.02018v1.pdf, page 8

[7] arXiv 2611.02018v1.pdf, page 16

[8] arXiv 2611.02018v1.pdf, page 24